Electrician's guide: the studio

Everything the electrician needs to wire one container studio: the first fix before the walls close, the second fix, the tests and the certificate.

Status: revision A2, 2026-10-28, PRELIMINARY. Every rating is design intent until a RURA-licensed electrician (class A) designs, installs and certifies the installation to RS 565-1:2011.

The two studios are wired the same, so one guide serves both.

Kinyarwanda: soma aya mabwiriza mu Kinyarwanda · Print: this guide as a PDF (not printed yet)

Words used here

- **End A**: the closed end wall, by the bed. **End B**: the cargo-door end, by the bathroom.
- **Back wall**: the long wall with no openings. **Front wall**: the long wall with the sliding glass door.
- Positions in the points table are millimetres from the finished end-A wall (x), from the finished back wall (y) and up from the finished floor (z).

The studio at a glance

What	Design
Supply	one inlet 32 A, IP67, IEC 60309 blue (2P+E), outside the back wall straight behind the board; any 230 V source
Main	32 A double-pole breaker (C curve) and a DIN kWh check meter; tails 6 mm² through a gland
Board	24 modules in two rows of 12: main, check meter, type 2 SPD, priority relay and its contactor, one RCBO (30 mA, type A) per circuit
Circuits	8, C1 to C8 (table below)
Maximum demand	30.0 A diversified: lights 0.3 + fans 0.3 + sockets 16.4 + the larger of the hob 11.6 or the heater 8.7 with the AC 4.3
Earthing	TT: one earth rod per container; never a combined neutral-earth (PME) supply
Cable route	20 mm conduit in the batten cavities, buried in spray foam after the first fix

When	What	Step
After the battens (S03)	First fix: earth rod, conduits, boxes, cables	S04
After the lining and the kitchen	Second fix: board, devices, tests, certificate	S19

Three hold points

1. **Battens first.** Start the first fix only when the battens (S03) are fixed and the floor sheets are down.
2. **Before the foam.** Every conduit, box and cable is in, insulation-tested, photographed wall by wall and signed before the spray foam (S05). The foam buries the conduits: nothing can be added later except by pulling through them.
3. **Before power-on.** The tests pass and the RURA certificate is signed before the inlet is ever fed.

S04: first fix

1. Set the board box on the back wall at the galley, and the inlet outside straight behind it: positions DB and PI in the points table.
2. Drive the earth rod (E1) outside end B and run its conductor to the board's earth bar; measure its resistance later, at the tests.
3. Run 20 mm conduit up from the board into the ceiling void, along it, then straight down to each box. No diagonal runs; fix every box to a batten.
4. Pull the cables per circuit: 1.5 mm² for C1 and C6, 2.5 mm² for C2, C3, C5, C7 and C8, 4 mm² for C4.
5. Take C6, the outdoor lights, out through one gland in the solid front strip at the bed end, then in UV-stable conduit outside.
6. Bathroom: an IP65 light and an IPX4 extract fan (zone 1, above the tray and below 2.25 m), no sockets, switched from outside the door.
7. Write each point's id on the steel beside its box, so the carpenter cuts the lining panels in the right places.
8. Test continuity and insulation resistance on every cable, photograph each wall, and only then sign the release for the foam.

Circuits: cable, breaker, run length and voltage drop · download CSV

circuit	serves	device	cable_mm2	design_A	derated_Iz_A	run_m	
C1	Indoor lights and fans	B6	1.5	0.6	13.6	34.9	
C2	Sockets: bed, sofa and deck	B16	2.5	8	18.3	24.3	
C3	Kitchen sockets, fridge and hood	B16	2.5	14	18.3	12.3	
C4	Induction hob	B20	4	15.2	24.4	4.4	
C5	30 L water heater	B16	2.5	8.7	18.3	3.9	
C6	Outdoor lights: deck, stair and roof	B6	1.5	0.2	13.6	16.5	
C7	AC provision	B16	2.5	12	18.3	3.9	
C8	Spare	B16	2.5		18.3		

Points: what, where and how high · download CSV

point	kind	circuit	description	x_mm	y_mm	z_mm	watts	face
DB	db		Unit board, 24 modules (2 x 12), by the galley	3678	0	1625	0	+y

point	kind	circuit	description	x_mm	y_mm	z_mm	watts	face
PI	inlet		Power inlet 32 A IP67 (IEC 60309 blue, 2P+E), outside behind the board	3678	-60	1225	0	-y
L1	light	C1	Ceiling light, bed	800	1126	2316	12	-z
L2	light	C1	Ceiling light, living	2300	1126	2316	12	-z
L3	light	C1	Ceiling light, galley	3778	1126	2316	12	-z
L4	light	C1	Bathroom light IP65	5223	850	2316	12	-z
L5	light	C1	Wall reading light over the bed (wall side)	300	0	1150	4	+y
L7	light	C1	Under-cabinet LED strip	4278	800	1790	10	-z
S1	switch	C1	Entry: bed (2-way), living and galley lights	4480	2252	1100	0	-y
S2	switch	C1	Bed: ceiling light (2-way), over the bedside table	1810	0	1000	0	+y
S3	switch	C1	Bathroom light + fan (outside)	5368	1770	1100	0	+y
F1	fan	C1	Bathroom extract 100 mm, IPX4 for zone 1, with run-on	4898	0	2100	20	+y

point	kind	circuit	description	x_mm	y_mm	z_mm	watts	face
F2	fan	C1	Wall fan 300 mm	1800	0	2050	50	+y
P1	usb	C2	Socket + USB-C in the headboard (wall side)	300	18	750	200	+y
P2	usb	C2	Bedside socket + USB-C (reading lamp)	1810	0	650	200	+y
P3	usb	C2	Socket + USB-C over the sofa arm, for the sofa and the table	3150	0	850	500	+y
P5	socket	C2	Deck socket IP66, outside by the slider	4680	2352	400	300	+y
P6	socket	C3	Worktop double socket 1	4578	275	1100	2000	-x
P7	socket	C3	Worktop double socket 2 (back wall)	4278	0	1100	1000	+y
P8	socket	C3	Fridge socket	4578	275	250	100	-x
P12	socket	C3	Double socket above the hood: hood + Wi-Fi router	4578	850	1720	120	-x

point	kind	circuit	description	x_mm	y_mm	z_mm	watts	face
P9	outlet	C4	Hob outlet plate, isolated at its RCBO on the board	4578	850	600	3500	-x
P10	outlet	C5	Heater DP switch + timer	4578	1540	1700	2000	-x
L8	light	C6	Deck light IP65, under the veranda canopy	2900	2395	1984	10	-z
S5	switch	C6	Deck light, beside the entry switch	4575	2252	1100	0	-y
S4	switch	C6	Stair and roof lights (2-way), IP65 at the stair foot	-200	2925	1100	0	+x
L10	light	C6	Stair light IP65	-900	2925	1500	8	+y
S6	switch	C6	Stair and roof lights (2-way), IP65 at the stair head	-105	2195	3579	0	-x
L9	light	C6	Roof terrace lights IP65 (x2), under the front top rail	1500	2285	3579	20	-z
P11	outlet	C7	AC provision: isolator	900	0	2150	1000	+y
AL1	alarm		Smoke alarm, 10-year battery	1550	1126	2316	0	-z

point	kind	circuit	description	x_mm	y_mm	z_mm	watts	face
AL2	alarm		Heat alarm, kitchen, 10-year battery	3428	1689	2316	0	-z
E1	earth		Earth rod 1.5 m copper-bonded in an inspection pit, behind the board	3678	-450	-700	0	+z

S19: second fix

1. Fit the board: the main breaker and check meter, the SPD, the priority relay and contactor, and the RCBOs (B6 for C1 and C6, B16 for C2, C3, C5, C7 and C8, B20 for C4).
2. Fit the devices: lights, fans, sockets, the hob outlet (P9, isolated at its RCBO on the board, in sight), the smoke and heat alarms (AL1, AL2).
3. Bond the container body, the front deck, the stair and the roof steelwork to the main earthing terminal.
4. Check the priority relay: with the hob on, the water heater (C5) and the AC (C7) must drop out.

Tests before power-on

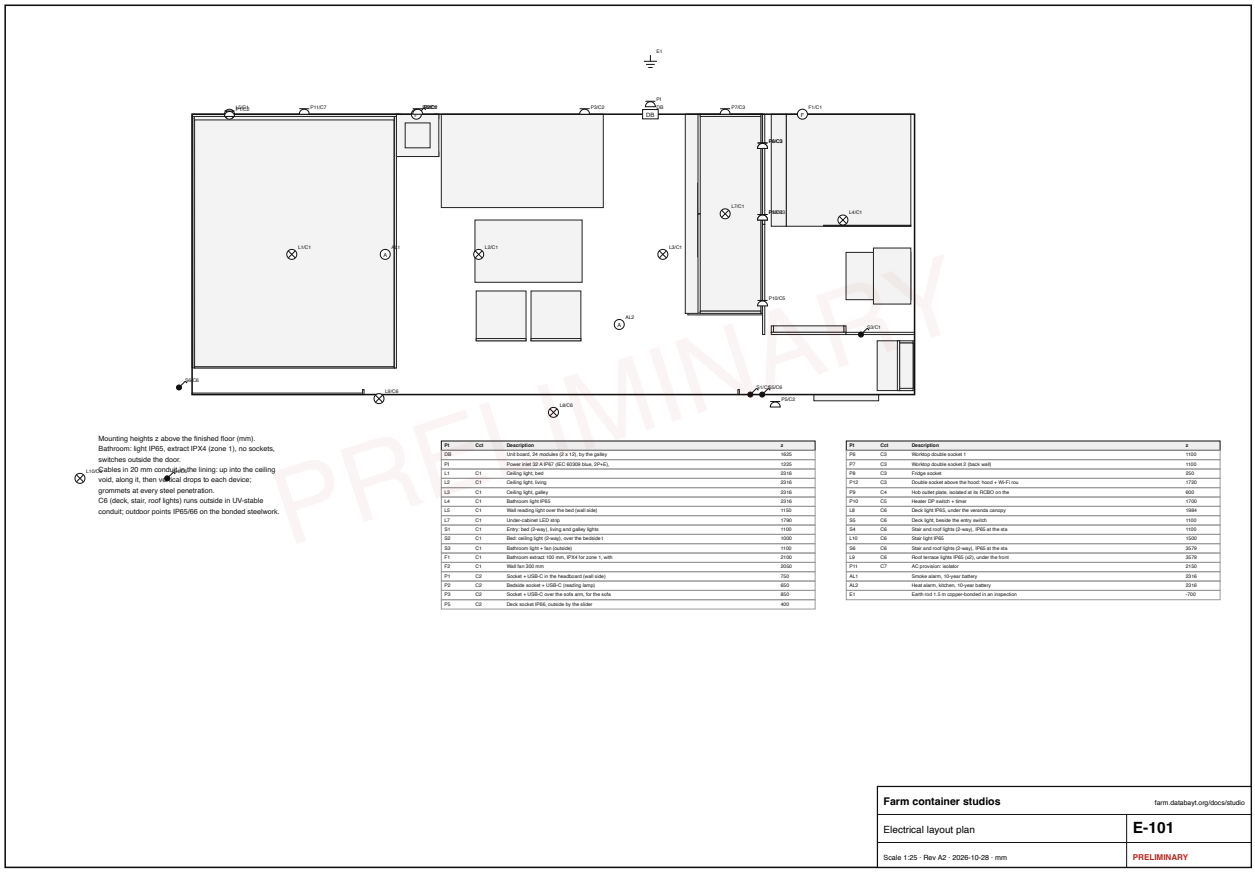
1. Continuity of every protective conductor.
2. Insulation resistance of every circuit.
3. Polarity at every socket and light.
4. Earth-electrode resistance of E1: write the value down.
5. The trip time of every RCBO.
6. The SPD status window, and the priority relay once more.
7. The RURA certificate, before the inlet is fed.

Feeding the inlet

- Any 230 V, 50 Hz source can feed the inlet (the site board, a generator or a solar inverter) if it protects its own cable at 32 A or less. Which source the farm uses is the `power guide`'s question.

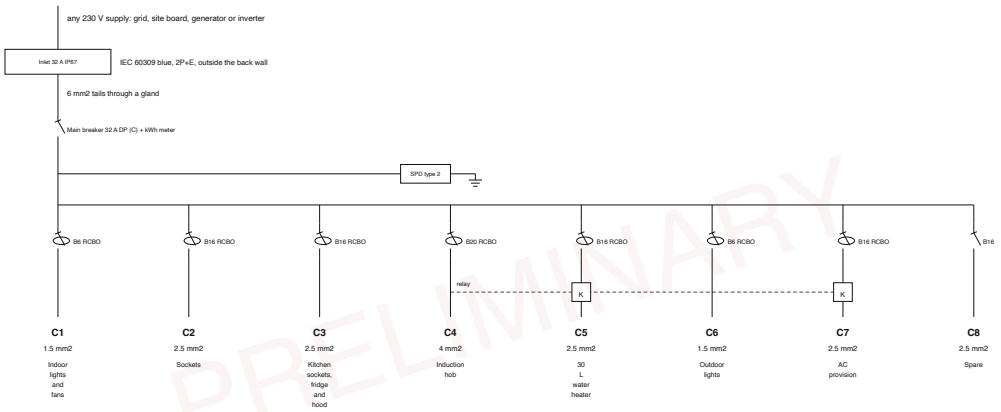
- The kWh meter in each studio is a check meter for monitoring only: electricity is in the room rate.

Drawings



E-101 Electrical layout plan · open full size

ONE STUDIO, FROM ITS INLET TO EVERY CIRCUIT



PRELIMINARY

The studio plugs in: any 230 V source protected at 32 A or less; the 32 A main breaker protects the inlet and tails whatever feeds them.
Priority relay on C4: while the hob runs, contactor K sheds C5 and C7 (heater and AC).
Diversified maximum demand about 30 A (AS-19), within the 32 A inlet. One RCBO 30 mA type A per circuit.
TT earthing: the studio's own earth rod at the board; never rely on the supply's earth (STU-37).
Ratings are design intent: a RURA class A electrician designs and certifies (STU-15), to RS 565-1:2011 (STU-14).

Farm container studios	farm.databayt.org/docs/studio
Single-line diagram: inlet and unit board	**E-201**
Scale NTS · Rev A2 · 2026-10-28 · mm	PRELIMINARY

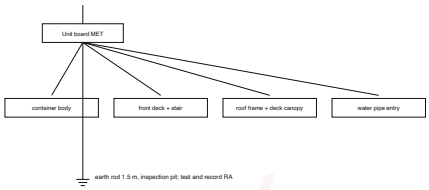
E-201 Single-line diagram: inlet and unit board · open full size

CIRCUIT SCHEDULE (Ib design current, Iz derated capacity, drop at the far point)

Cct	Serves	Device	mm2	Ib A	Iz A	Run m	Drop %
C1	Indoor lights and fans	B6	1.5	0.6	13.6	34.9	0.25
C2	Sockets: bed, sofa and deck	B16	2.5	8.0	18.3	24.3	1.52
C3	Kitchen sockets, fridge and hood	B16	2.5	14.0	18.3	12.3	1.35
C4	Induction hob	B20	4	15.2	24.4	4.4	0.32
C5	30 L water heater	B16	2.5	8.7	18.3	3.9	0.27
C6	Outdoor lights: deck, stair and roof	B6	1.5	0.2	13.6	16.5	0.03
C7	AC provision	B16	2.5	12.0	18.3	3.9	0.37
C8	Spare	B16	2.5	-	-	-	-

Maximum demand item	A
lighting	0.3
fans	0.3
sockets	16.4
hob	11.6
heater	8.7
ac	4.3
total_A	30.0

EARTHING AND BONDING (TT)



earth rod 1.5 m, inspection pit; test and record RA

Bonding: finely stranded green-yellow (16 mm2 [A]) from each steel item to the MET. RA x 30 mA must stay under 50 V; aim much lower for lightning (measure on site). Tests before handover: continuity, insulation resistance, polarity, earth electrode resistance, RCD trip time, SPD status; certificate from the licensed electrician. Bathroom: zone 1 above the tray to 2.25 m (IPX4); 30 mA on every circuit (STU-38).

PRELIMINARY

Farm container studios	farm.databayt.org/docs/studio
Circuit schedule, earthing and bonding	E-202
Scale NTS · Rev A2 · 2026-10-28 · mm	PRELIMINARY

E-202 Circuit schedule, earthing and bonding · open full size